

Show all your work in the space provided for each problem. All problems will be graded for correctness, completeness, conciseness, and clarity. Do your best. Don't cut corners.

1. (9 pts) Use the given dictionary to translate each English sentence into sentential logic.

B: A sequence is bounded.

C: A sequence converges.

D: A sequence diverges.

M: A sequence is monotone.

a) A sequence diverges when it is not monotone.

$$(\sim M) \rightarrow D$$

b) If a sequence is bounded and monotone it will converge.

$$(B \wedge M) \rightarrow C$$

c) A divergent sequence is unbounded but is monotone

$$D \rightarrow ((\sim B) \wedge M)$$

2. (6 pts) Write DeMorgan's Laws in sentential logic **and** in set theory using proper notation.

$$\sim (P \vee Q) \equiv (\sim P) \wedge (\sim Q)$$

$$\sim (P \wedge Q) \equiv (\sim P) \vee (\sim Q)$$

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

3. (9 pts) a) Fill in the truth table for the $p \leftrightarrow q$.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

b) Give a statement using only the connectives $\wedge$ and $\rightarrow$ that is logically equivalent to $p \leftrightarrow q$

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

c) Use disjunctive normal form to write a statement equivalent to $p \leftrightarrow q$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\sim p \wedge \sim q)$$

4. (9 pts) Pick the best predicate logic representation for each English sentence. Use the dictionary: $g(x)$ means x is a group $a(x)$ means x is associative $c(x)$ means x is commutative

a) Every group is associative.

- i) $\forall x[g(x) \wedge a(x)]$ ii) $\exists x[g(x) \wedge a(x)]$ iii) $\forall x[g(x) \rightarrow a(x)]$

b) Some group is commutative.

- i) $\forall x[g(x) \wedge c(x)]$ ii) $\exists x[g(x) \wedge c(x)]$ iii) $\exists x[g(x) \rightarrow c(x)]$

c) Some group is neither associative nor commutative

- i) $\exists x[g(x) \wedge (\sim a(x)) \wedge (\sim c(x))]$ ii) $\exists x[g(x) \rightarrow \{(\sim a(x)) \vee (\sim c(x))\}]$
 iii) $\exists x[g(x) \vee (\sim a(x)) \vee (\sim c(x))]$

5. (8 pts) Write each statement in the form If _____, then _____.

a) Every prime larger than 2 is odd.

If x is both prime and $x > 2$, then x is odd

b) The function f is even when $f(-x) = f(x)$ for every x in the domain.

If $f(-x) = f(x)$ for every x in the domain, then f is even

6. (10 pts) Consider the argument $A \vee (\sim C), B \wedge E, B \rightarrow (C \wedge D) \therefore A$. Using known rules of deduction prove the argument is valid. Label each step using the proper name for the rules of deduction used.

- ① $B \wedge E$ premise
- ② B conjunctive simplification ①
- ③ $B \rightarrow (C \wedge D)$ Premise
- ④ $C \wedge D$ ②, ③ modus Ponens
- ⑤ C ④ conj. simp.
- ⑥ $A \vee (\sim C)$ Premise
- ⑦ $\sim(\sim C)$ ⑤ Double negation
- ⑧ A disjunctive syllogism ⑥, ⑦

7. (9 pts) Consider the statement: If a is even and b is even then $a+b$ is even.

a) Write the converse of this statement.

If $a+b$ is even, then a is even and b is even.

b) Write the inverse of this statement.

If it is not the case that both a and b are even then $a+b$ is not even.
If either a is odd or b is odd then $a+b$ is odd.

c) Write the contrapositive of this statement.

If $a+b$ is odd then either a is odd or b is odd.

8. (6 pts) **Disprove** the following statement.

For all real numbers $x > 1$ and $y > 1$, $xy \geq x+y$.

Disprove the universal statement with a counterexample. Find real numbers $x > 1$ and $y > 1$ with $xy < x+y$.

Let $x = \frac{3}{2}$ and $y = 2$. Both $x > 1$ and $y > 1$

But $x \cdot y = 3$ and $x + y = 3.5$

so $xy \neq x+y$

9. (10 pts) Prove the following statement.

If x is rational and y is rational then $x + y$ is rational

Direct Proof

Let x and y be rational

$x = \frac{a}{b}$ and $y = \frac{c}{d}$ where a, b, c, d are integers and b, d are not zero.

Then $x + y = \frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$ is rational since

$ad + bc \in \mathbb{Z}$ whenever $a, b, c, d \in \mathbb{Z}$ and

$bd \neq 0$ by the zero product property of real numbers
when $b \neq 0$ and $d \neq 0$.



10. (6 pts) What statement logically equivalent to $p \rightarrow q$ is used in a proof by contrapositive?

$$p \rightarrow q \equiv (\neg q) \rightarrow (\neg p)$$

What statement logically equivalent to $p \rightarrow q$ is used in a proof by contradiction?

$$p \rightarrow q \equiv (p \wedge (\neg q)) \rightarrow (r \wedge (\neg r))$$

11. (8 pts) Prove that the statement $(p \wedge q) \vee r$ is logically equivalent to $(p \vee r) \wedge (q \vee r)$

p	q	r	$p \wedge q$	$(p \wedge q) \vee r$	$p \vee r$	$q \vee r$	$(p \vee r) \wedge (q \vee r)$
T	T	T	T	T	T	T	T
T	T	F	T	T	T	F	F
T	F	T	F	T	T	T	T
T	F	F	F	F	T	F	F
F	T	T	F	T	F	T	F
F	T	F	F	F	F	F	F
F	F	T	F	T	F	T	F
F	F	F	F	F	F	F	F

Explain:

Since the two columns are identical in every row the statements are equivalent.

12. (6 pts) Let $A = \{2, 4\}$, $B = \{1, 3, 5, 7\}$, and $C = \{2, 5, 6\}$ be subsets of the first 9 positive integers. List the elements in each of the following sets.

a) $(A \cup C) \setminus B = \{2, 4, 5, 6\} \setminus \{1, 3, 5, 7\} = \{2, 4, 6\}$

b) $C \cap B^c = \{2, 5, 6\} \cap \{2, 4, 6, 8, 9\} = \{2, 6\}$

c) $A \times C = \{(2, 2), (2, 5), (2, 6), (4, 2), (4, 5), (4, 6)\}$

13. (4 pts) Let A be the set of all multiples of 4 and B the set of all multiples of 6. Describe all the elements of $A \cap B$.

$$\begin{aligned}
 A \cap B &= \text{set of all multiples of 4 and 6} \\
 &= \text{set of all multiples of 12} \\
 &= \{12 \cdot k : k \in \mathbb{Z}\}
 \end{aligned}$$